SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Operating Systems
COURSE CODE: CSA04

COURSE OUTCOME COVERED

CO3: Evaluate memory management mechanisms such as linking, paging, and segmentation to determine their effectiveness for different system requirements

Assessment Tool Weightage: 30%

QUESTIONS: 5 Total Marks: 50

CASE STUDY

1. An embedded system has limited physical memory (64 MB) and runs several small applications. Each application consists of multiple modules, and memory usage must be minimized. The system should load only the required modules when necessary.

   Tasks:

   a. Evaluate whether static linking or dynamic linking would be more suitable for this system.

   b. Compare the memory and storage requirements of the two approaches. c. Identify one disadvantage of your selected approach.

   d. Justify your final choice based on the requirements of the embedded system.

2. A cloud server runs hundreds of processes simultaneously. Processes have different memory requirements and may start or terminate at any time. The system should minimize external fragmentation and efficiently utilize available physical memory. **Tasks:**

   a. Evaluate contiguous memory allocation and paging for this system. b. Determine which mechanism would provide better memory utilization. c. Explain how the selected mechanism handles processes that require non contiguous memory.

   d. Justify your choice based on fragmentation, memory utilization, and process management.

3. A software application consists of several logically independent components such as:

   • Code

- Global data
- Stack
- Heap
- Shared libraries

The developers want memory protection between these components and want different access permissions for each logical section.

Tasks:

a. Evaluate whether paging or segmentation is more suitable for this application.

b. Explain how the selected technique can provide protection and controlled access.

c. Identify one limitation of the selected technique.

d. Justify your choice based on logical organization, protection, and memory utilization.

4. A modern operating system must support:

- Large applications
- Multiple processes
- Efficient physical memory utilization
- Protection between processes
- Virtual memory
- Programs larger than available physical memory

The system designers are considering paging, segmentation, and segmentation with paging.

Tasks:

a. Compare the three memory management mechanisms for the given requirements.

b. Evaluate which mechanism provides the best combination of protection, flexibility, and memory utilization.

c. Explain how your selected mechanism handles logical memory and physical memory.

d. Justify why the other two mechanisms are comparatively less suitable.

5. Consider two systems:

System A:

A high-performance gaming computer with large RAM, where applications require fast memory access and frequently load large executable files.

System B:

A multi-user server running many independent processes with varying memory requirements.

The OS designers are considering contiguous allocation, paging, and segmentation. **Tasks:**

a. Evaluate the suitability of each memory management technique for System A and System B.

b. Select the most appropriate mechanism for each system.

c. Compare your choices in terms of:

   a. Fragmentation
   b. Memory utilization
   c. Protection
   d. Address translation overhead
   e. Flexibility

d. Provide a final recommendation for both systems with appropriate

justification. **RUBRICS FOR CALCULATION**

Criteria	**Weightage**	**Level 4 – Exemplary (5 Marks)**	**Level 3 – Proficient (4 Marks)**	**Level 2 – Developing (2–3 Marks)**	**Level 1 – Beginning (0–1 Mark)**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

1. Embedded System - Static Linking vs Dynamic Linking.

a) Static Linking technique :-

Dynamic linking is more suitable for the embedded system.

The system has only 64MB of physical memory and runs several small applications each application has multiple modules, but all modules are not required at the same time. Dynamic linkage allows required modules or libraries to be loaded only when they are needed.

b) Memory and storage comparison:

Feature	Static linking	Dynamic linking.
Library code	included in executable	Stored seperately
executable size	Large	Small
Memory usage	Higher	Lower.
Storage	More	Less
Sharing	Limited	Libraries can be shared
Loading	During linking	during the execution.

For example, if several application use the same library, static linking may include a seperate copy in every application. Dynamic linking can keep one shared copy in memory.

c) Disadvantage:

The disadvantage of dynamic linking is runtime dependency. If the required library is missing, Corrupted, or incompatible, the application may fail to executed. It may also introduce a small runtime loading overhead.

d) Justification:

Dynamic linking is preferred because:

* Physical Memory is limited to 64 MB.
* it reduces executable size
* common module can be shared.
* it reduce duplicate code.
* it saves both Memory and storage.

Dynamic linkage is the best choice because it minimizes memory usage and allows module to be loaded and shared when required.

2) Cloud server - contiguous allocation vs pagging.

a) Evaluation:

In contiguous memory allocation, each process must occupy a single continuous block of physical Memory.

Process A
Process B
Free space
Process C

When process terminate, free spaces may become scatterd, resulting in external fragmentation.

In paging, logical memory is divided into fixed size pages and physical memory is divided into frames.

Page 0 → Frame 5
Page 1 → Frame 2
Page 2 → Frame 8
Page 3 → Frame 1

b) Better memory utilization:-

Paging provides better memory utilization because any available frame can be allocated to a page.

c) Handling non-contiguous Memory:-

Paging allows different pages of the same process to reside in different physical frames. A pages table stores the mapping b/w logical pages and physical frames.

d) Justification:-

Paging is better for the cloud server because:-

Hundreds of process may execute simultaneously.

Processes have different memory requirements.

Processes can start and terminate at any time.

External fragmentation is eliminated.

Non-contiguous memory allocation is supported.

Available physical memory is used efficiently.

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3) Application components - paging vs segmentation:-

The application contains;

* code.
* Global data.
* Stack.
* Heap.
* Shared libraries.

a) Suitable technique:-

Segmentation is more suitable because the application is naturally divided into logically independent sections.

Each section can be represented as a separate segment.

Program
- Code
- Global data
- Stack
- Heap
- Shared libraries.

b) Protection and controlled access:-

each segment can have its own base address, limit, and access permission.

Segment	Permission.
Code	Read + execute
Global data	Read + write
Stack	Read + write
Heap	Read + write
Library	Read + execute.

c) Limitation:

The main limitation is external fragmentation because segments have variable sizes. Free memory can become divided into small unusable holes.

b) Justification :-

It represents the logical structure of program.

Different sections can have different permissions.

code can be protected from modification.

Data, stack and heap can be independently protected.

it provides better logical organization.

Segmentation is the better choice because the application has clearly defined logical components requiring different protection and access permissions.

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4. Modern operating system - paging, segmentation and seg. with paging.

a) comparison :-

Feature	paging	segmentation
logic organization	Moderate	Excellent
Memory utilization	High	Moderate
External fragmentation	NO	yes
Protection.	Good	Excellent
virtual Memory	Excellent	good
Flexibility	High	High.

b) Best Mechanism :-

⇒ Segmentation with paging provides the best combinates of ;

* Protection
* flexibility

* logical organization.
* Efficient physical memory utilization.
* Virtual Memory.

c) Logical and physical memory handling:-

In segmentation with paging, the logical address is first divided into a segment number and offset.

The segment is then divided into pages. The page number is mapped to physical fram using the page table.

Logical Address
↓
Segment Table
↓
Page Number
↓
Page table
↓
Physical Frame.

d) Why the other two are less suitable:-

Paging alone: Provides excellent memory utilization and virtual memory but does not naturally represent logical program sections.

Segmentation alone: Provides excellent logical organization and protection but suffers from external fragmentation.

Segmentation with paging: combines the advantages of both.

5) System A and System B

System A - High-performance Gaming computer.

a) Evaluation of techniques :-

Contiguous allocation :-

Simple implementation.

Very fast address translation.

Low overhead.

Can cause external fragmentation.

Paging :-

Efficient Memory utilization.

Supports non-contiguous allocation.

Suitables for virtual Memory.

requires page-table management.

Segmentation:

Provides logical organization.

Supports protection.

Can cause external fragmentation.

b) Selection for System A:

for the given scenario, contiguous allocation can be selected because System A has large RAM and emphasizes fast Memory access.

System B - Multi-user environment:-

a) Evaluation:

Contiguous allocation can cause external fragmentation when processes start and terminate. Segmentation also suffers from external fragmentation because segments have variable sizes.

Paging allows processes to occupy non-contiguous physical frames, making it more suitable.

b) Selection:

Paging is the most appropriate mechanism for System B.

c) Comparison:

Factor	Contiguous	Paging	Segmentation.
Fragmentation	High external	Low external	High
Memory utilization	Moderate	High	Moderate
Protection	Moderate	Good	excellent
Translation overhead.	Low	Moderate	Moderate
Flexibility.	Low	High	High.

d) Final recommendation:-

System A → contiguous allocation.

fast, low overhead, suitable for large memory.

System B → Paging:

Better memory utilization.